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METHOD OF STATEMENT FOR CHILLED WATER PIPE PRESSURE TEST

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1.0 Objective

To define the controlled, systematic, and safe procedure for conducting the hydrostatic pressure test on the Chilled Water (CHW) piping system. The primary objective is to empirically verify the mechanical integrity, joint soundness, and overall leak-tightness of the installed, ensuring its fitness for purpose and full compliance with the project's quality and safety standards.

2.0 Purpose

This document serves as the single-source-of-truth for all personnel involved in the testing works. Its purpose is to:

- **Chilled Water Piping System:** All newly installed chilled water supply and return piping, including all associated fittings, valves, and connections, from the chiller plant to the fan coil units (FCUs) and air handling units (AHUs).
- Provide a clear, step-by-step methodology to eliminate ambiguity and ensure consistent, repeatable results.
- Establish definitive, measurable acceptance criteria for a successful test.
- Integrate WSH best practices at every stage to ensure the highest level of safety.
- Form the basis for formal inspection, witnessing activities, and obtaining official approval from the Client, Consultant and relevant authorities.

3.0 Scope of Work

This procedure governs all labour, materials, and equipment required for the hydrostatic pressure testing of the newly installed Chilled Water piping network.

Inclusions:

- All CHW supply and return main lines, headers, and branch lines.
- All piping materials (e.g., Carbon Steel, Stainless Steel).
- All types of joints within the test boundary, including welded, flanged, and grooved connections.
- The scope extends from the primary isolation points in the chiller plant room to the final isolation valves immediately before all terminal equipment (AHUs, FCUs).

Exclusions from Test Pressure:

The test pressure will **not** be applied to equipment or components not rated for it. These items will be positively isolated from the test section via closed valves or the installation of temporary blank flanges:

- Chillers, Pumps, Heat Exchangers.
- Air Handling Unit (AHU) and Fan Coil Unit (FCU) coils.
- Automatic Control Valves (e.g., PICV, DRV), Strainers, and Expansion Vessels.
- Pressure Relief Valves and permanent instrumentation.

4.0 Responsibility

Role	Key Responsibilities
Project Manager	Overall accountability for successful project execution. Ensures allocation of necessary resources (manpower, calibrated equipment) and acts as the primary liaison with the client for scheduling and issue resolution.
Site Engineer	On-site authority for the test. Manages the Permit-to-Work, supervises the technical team, verifies all pre-test checks are complete, records all test data, and formally requests client/consultant witness inspections.
Site Supervisor	Independent quality authority. Verifies material compliance and the validity of calibration certificates. Witnesses the test, validates results against acceptance criteria, and is responsible for compiling and submitting the final, comprehensive
WSH Supervisor	Enforces 100% compliance with the approved Risk Assessment (RA) and Safe Work Procedure (SWP). Conducts pre-work toolbox talks, ensures the test area is correctly secured, and has the authority to stop work if any unsafe condition arises.
Skilled Technicians	Execute the physical work under direct supervision, including system filling, venting, pressurization, and depressurization. Responsible for the correct installation of test equipment and diligent monitoring.

5.0 Preparation (Pre-Test)

A formal "Request for Inspection" (RFI) will be submitted to the client/consultant 24 hours prior to the planned test. No test will proceed without the following preparations being complete and verified by the Site Engineer or Supervisor

5.1 System & Documentation:

- A fully approved **Permit-to-Work (PTW)** for pressure testing is mandatory.
- The approved construction drawings shall be marked up to clearly delineate the exact boundaries of the pipe section to be tested. This drawing will be attached to the test record.
- Verification that all welding, fabrication, and permanent pipe support installations are 100% complete and have passed visual inspection.

5.2 Isolation & Equipment Protection:

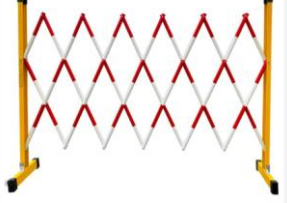
- The test section shall be physically and positively isolated. Where possible, **rated blank flanges/spades** will be used in preference to relying solely on valves.
- A final walk-down shall be conducted to ensure all excluded components are confirmed to be isolated from the section under test.
- Confirm that vents are installed at all system high points and drains are available at all low points

5.3 Test Equipment & Calibration:

- A minimum of **two (2) pressure gauges** shall be used for each test.
- All pressure gauges must have a valid calibration certificate from a **accredited laboratory**, with the calibration date.
- The hydrostatic test pump must be in excellent working condition, equipped with a functional non-return valve and a certified pressure rating.

6.0 Photos of Hand Tools & Materials Relevant

Category	Item Description
Test Equipment	1. Calibrated, liquid-filled Pressure Gauges with valid stickers. 2. Manual Hydrostatic Test Pump. 3. High-pressure rated flexible hose assembly.
Isolation Gear	1. New, correct-sized Gaskets. 2. Torque Wrench with socket set.
Safety Equipment	1. Safety Barriers. 2. Laminated, Warning Signs. 3. Full-body harnesses and lanyards for any work at height. 4. Ladder, Scaffold, scissor lift if necessary
PPE	1. Safety Helmets. 2. Safety Goggles. 3. Safety Boots. 4. Hand Gloves.

 Pressure Gauges	 Manual Hydrostatic Test Pump.	 High-pressure rated flexible hose assembly
 Gasket	 Torque Wrench with socket set.	 Safety Barriers.



7.0 Testing Method and Commissioning Procedures

Step-by-Step Execution Protocol:

1. **Water Filling & Air Purging:** The section will be slowly filled with clean water from a designated low point. All high-point vents will be opened. Filling will continue until a solid, air-free stream of water is discharged from all vents, which will then be closed. This step is critical and will be performed with diligence.
2. **Incremental Pressurization:** Pressure will be increased in controlled stages:
 - **Stage 1 (50%):** Raise pressure to 50% of the final test pressure. Hold for 10 minutes for an initial stabilization and walk-down to check for obvious leaks.
 - **Stage 2 (100%):** If Stage 1 is clear, continue to raise pressure slowly to the full required test pressure.
3. **Stabilization & Formal Test Start:**
Once at 100% test pressure, isolate the pump. Allow the system to stabilize for **30 minutes**. After stabilization, the formal test begins. The start time and pressure on both gauges will be recorded and formally witnessed.
4. **24-Hour Holding Period:**
The system shall hold the full test pressure for a minimum duration of **24 hours**. A comprehensive visual inspection of every single joint and connection will be conducted during this period.
5. **Test Conclusion & Acceptance:**
The test is declared **SUCCESSFUL** only if **both** of the following criteria are met:
 - **Criterion A:** There is **ZERO visible leakage** (no drips, moisture, or weeping) from any pipe, fitting, or joint.
 - **Criterion B:** There is **ZERO pressure drop** from the initial witnessed reading on the calibrated gauges after the 24-hour period.
6. **Depressurization:** Upon successful sign-off, pressure will be released in a slow, controlled manner by opening a drain valve.

8.0 Event of Test Failure Re-Testing Procedure

1. A failed test necessitates a complete re-test of the section. **The system will not be simply "topped up" to pressure.**
2. The entire hydrostatic test procedure, as detailed in Section 7.0, will be repeated **from the beginning**. This includes:
 - Refilling the section with water.
 - Thoroughly purging all air.

- Incremental pressurization.
 - Commencement of a **new, full 24-hour holding period**.
3. A new Pressure Test Certificate shall be initiated for the re-test, referencing the original failed test's NCR number for full traceability.

9.0 Documents Handover

Upon successful completion, a formal **Hydrostatic Test Pack** will be compiled and submitted for each tested system.

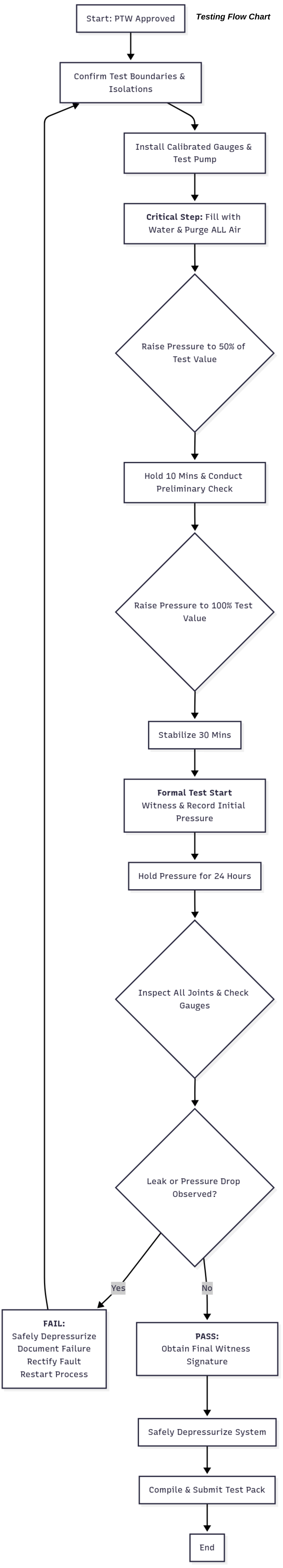
Document Control Cover Sheet.

1. **Completed & Signed Pressure Test Certificate**, detailing location, test pressure, medium, duration, and final result. It must be signed by the Main Con M&E, RTO Inspector, and the Client's Representative.
2. **The Marked-up "As-Built" Drawing** clearly highlighting the successfully tested section.
3. Copies of the **Calibration Certificates** for the specific pressure gauges used.
4. The final, closed-out **Permit-to-Work (PTW)**.

10.0 Safety & Health

The safety of all personnel is the paramount consideration.

- **Risk Assessment (RA):** All personnel involved must attend a specific **Toolbox Talk** before work commences, where the RA for pressure testing will be reviewed.
- **Exclusion Zone:** The test area will be demarcated with **hard barriers** and prominent, warning signs. No personnel are permitted to enter the exclusion zone without authorization from the supervisor in charge.
- **Stored Energy Hazard:** All personnel will be instructed that a pressurized system contains immense stored energy. **It is strictly forbidden to strike, tighten, or perform any work on the system while it is under pressure.**
- **Emergency Plan:** The location of the nearest first aid station and emergency contact numbers will be displayed at the test area. All team members will be briefed on the safe emergency depressurization procedure.



**PRESSURE TEST REPORT**

PROJECT NAME			
LOCATION		EQUIPMENT DESIGNATION	
AREA SERVED		EQUIPMENT MODEL	
SYSTEM		MANUFACTURER	

PIPE SIZE	
TYPE OF GAS / WATER	
INITIAL HOLDING PRESSURE (PSI/bar)	
TEST START ON DATE / TIME	
FINAL PRESSURE READING (PSI/bar)	
DURATION OF TEST (HOURS)	
TEST END ON DATE / TIME	

Comments: (If there is no additional comments enter NONE)

Tested By		Witnessed By	Verified By
Signature			
Name			
Designation			
Company			
Date			